

HABITAT-ORIENTED SAMPLING OF MACROINVERTEBRATES AFFECTS THE DETERMINATION OF ECOLOGICAL STATUS IN TEMPORARY MEDITERRANEAN RIVER SYSTEMS

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ABSTRACT

Inter-annual and annual variation in precipitation levels in Mediterranean temporary river systems strongly influence riverine flow regimes and as a result habitat availability for biological assemblages. Under-sampling of less well-presented microhabitats in such changeable, dynamic systems can result in information loss leading to misclassification of Water Framework Directive (WFD) compliant ecological status, with serious consequences for Programmes of Measures in River Basin Management Plans. This paper compares two benthic macroinvertebrate sampling protocols tested in intermittent Mediterranean streams ($n=40$) in the Algarve region of southern Portugal. The officially adopted WFD compliant European assessment system for the ecological quality of rivers using benthic macroinvertebrates, a composite sampling protocol, focuses sampling a greater area of the most representative microhabitats at a given sampling site (proportional habitat sampling). A non-proportional habitat sampling protocol was applied to test the assumption that taxa may have preference for different microhabitats and that this could influence classification of ecological status. Variation was detected in the levels of similarity in the structure and the composition of the benthic macroinvertebrate community within sampling sites and rivers, based on material collected using the two collection methods. Ecological index scores and biotic metrics were higher when the non-proportional oriented sampling protocol was utilized resulting in a 'good ecological status' classification at sites that had been classified as 'Bad' using the official WFD. Decision tree analysis results indicated that changes in the area of individual microhabitats sampled among protocols was the principal driver behind differences in ecological status derived. The results indicate that more realistic classifications could be achieved in intermittent Mediterranean rivers when considering and sampling less well-represented microhabitat types (non-proportional sampling) rather than distributing sampling effort by the proportion of the dominant habitat present as currently used for the WFD protocol. The application of the non-proportional microhabitat-based approach, which accounts for selective preferences of some benthic macroinvertebrate taxa, would ensure that a greater proportion of a given site community contributes to the assessment. Copyright © 2014 John Wiley & Sons, Ltd.

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INTRODUCTION

The Water Framework Directive (WFD, EC, 2000) is one of the most important and ambitious European legislative measures passed for the sustainable management of aquatic resources (Letcher and Giupponi, 2005). Outlining 'a framework for the protection of inland surface waters, transitional waters, coastal waters and groundwaters', the WFD promotes catchment level management of water resources and based on the typology of water body set in the context of the river basin. The WFD legislates for the development and implementation

of ecological assessment and classification systems using biological quality elements (benthic macroinvertebrates, fish and aquatic flora) and hydromorphological and physicochemical support elements to determine ecological status. Member states are expected to achieve 'good ecological status' for all surface water bodies by 2015. Accurate assessment of ecological status is the most forward looking but technically challenging part of the WFD because results directly contribute to the development of mandatory River Basin Management Plans and Programmes of Measures for water bodies that fall below good ecological status.

Rapid bioassessment protocols (RBPs) are widely used to assess the ecological status of wadeable streams and rivers (Plafkin *et al.*, 1989; Klemm *et al.*, 1990; Norris and Georges, 1993) and have been adapted to meet WFD

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requirements using standardized methodologies (AQEM, 2002; Hering *et al.*, 2003; Gabriels *et al.*, 2012; Tagliaferro *et al.*, 2013). The benthic macroinvertebrate fauna is perhaps the most widely studied freshwater biological element used in biomonitoring and RBP because the taxonomic and functional diversity and ubiquity of this community is well known and guarantees some detectable and measurable response to stressors (Rosenberg and Resh, 1993; USEPA, 2002).

The aim of biomonitoring sampling protocols, such as benthic macroinvertebrate sampling, is to obtain a representative sample of the target community at a given site using a repeatable, standardized methodology (Haase *et al.*, 2009; Gabriels *et al.*, 2012). Although sampling protocols differ in the type of sampling gear used, collection methods, sampling area, the number of replicates and sampling effort, the data obtained (structural or functional diversity, relative abundance, scores derived from the presence/absence or abundance of given indicator species) are used to calculate metrics that provide information, together with data on hydromorphological and physicochemical support elements, on ecosystem health.

The WFD requires member states to standardize sampling methods in order to present comparable results across Europe. This exercise has resulted in several large-scale inter-regional projects between 2000 and 2005 such as The Development and Testing of an Integrated Assessment System for the Ecological Quality of Streams and Rivers throughout Europe using Benthic Macroinvertebrates (AQEM) and Standardisation of River Classifications (STAR) and national projects such as the Spanish initiative GUADALMED (Jáimez-Cuéllar *et al.*, 2002; Prat, 2002). More recently, the MIRAGE project has focused on the development of a range of complementary tools to characterize different aspects of temporary Mediterranean rivers and streams globally for more effective assessment and management of these systems (Nikolaidis *et al.*, 2013). These projects have helped in the characterization of aquatic ecosystems, the development of sampling protocols, the standardization of methodologies and the inter-calibration of results (Hughes and Malmqvist, 2005; Feio *et al.*, 2013).

Most macroinvertebrate sampling protocols, including many developed for the WFD, are based on information taken from a composite sample rather than a sample based on the collection of individual replicates (Haase *et al.*, 2009; Gabriels *et al.*, 2012). The Portuguese WFD benthic macroinvertebrate sampling protocol for rivers (INAG, 2008), an adaptation of the AQEM protocol, comprises a composite sample of six 1 × 0.25 m subsamples collected with a hand net in proportion to the presence and extent of four inorganic and two organic microhabitat types in the sampling area (refer to Hering *et al.*, 2004). Similar to the

AQEM methodology described by Hering *et al.* (2004), this study applies the term microhabitat to mineral and biological substrates sampled. According to the EN ISO 27828 (1994), protocol sampling effort provides accurate characterization of overall benthic community, based on the percentage presence of each microhabitat sampled. However, a possible disadvantage of this method is that less well-represented microhabitats are under-sampled in proportion to the area they cover in relation to overall microhabitat, thereby omitting taxa that may display a strong preference for them. The resulting data loss could affect the final classification of ecological status. The consequences of misclassification of the ecological status of a water body (e.g. as 'moderate' rather than 'good') status could have serious socioeconomic consequences in the development and implementation of Programmes of Measures to meet 'Good' status by 2015 as part of River Basin Management Plans. The study of habitat selection and population interactions has a long tradition in ecology (Rosenzweig, 1991). Generally, the strongest separation of species by habitat selection is the result of distinct species preferences and inter-specific competition for resources. Numerous studies demonstrated that habitat characteristics influence factors such as growth rate, reproduction and species distribution within lotic or lentic habitat (Giller and Malmqvist, 1998; Thomaz and Cunha, 2010). We therefore hypothesize that non-proportional habitat-oriented sampling protocol could potentially influence ecological status and consequently the classification of water bodies.

This study compares results obtained from two benthic macroinvertebrate field sampling protocols and how they influence the classification of ecological status in intermittent Mediterranean river systems situated in the Algarve region of southern Portugal. It is known that inter-annual and annual variation in precipitation levels strongly influences flow regime, habitat availability and ecological processes in these systems (Gasith and Resh, 1999; Bonada *et al.*, 2008), affecting microhabitat availability. The two methods compared in this study, proportional habitat sampling (PHS) versus non-proportional habitat sampling (NPHS), employ identical sampling effort and area covered but vary in how sampling is distributed according to the microhabitats present. We aimed to assess the effect of sampling based on (i) habitat availability that complies with composite WFD mandatory sampling for Portuguese streams and rivers or (ii) habitat selectivity to test a new non-proportional habitat-oriented sampling protocol. Thus, the aim of this study was to evaluate how these different approaches influence indices and metrics derived and the implications for integrated water resource management planning under the WFD.

METHODS

Study area

The streams and rivers of the Hydrographical Administrative Region for the Algarve (ARH-Algarve), located in southern Portugal (Figure 1), are Mediterranean in character with a temporary flow regime, exhibiting marked but predictable natural cycles of flood and drought that vary in intensity according to annual and inter-annual rainfall levels (Gasith and Resh, 1999; Bêche and Resh, 2007a, 2007b). These patterns are directly related to different phases of the North Atlantic Oscillation (Trigo *et al.*, 2004; Hughes *et al.*, 2012). Maximum discharge occurs during the winter months (November to April), falling to virtually zero from late spring (May to early June) onwards. Many of the region's fluvial systems have little or no surface water flow during the late spring and summer months, although isolated summer pools that remain in the streambeds provide important refugia for aquatic communities.

Composite non-proportional versus proportional habitat sampling

A total of 40 sites were sampled in 2009, representing all WFD compliant monitoring sites established by the Portuguese Water Institute (INAG). In accordance with INAG (2008) benthic macroinvertebrate sampling protocol and because of intermittent or temporary character of most Algarve streams, field sampling was carried out in spring (April) under continuous flow conditions when community and habitat diversity are generally considered greatest. The 50-m-long sampling area was selected to cover the greatest possible diversity of microhabitats representative of the reach 25 m upstream and 25 m downstream of an area of turbulent flow (i.e. a riffle).

We tested two field sampling procedures: (i) the WFD compliant composite sampling procedure (from here on denoted as PHS) and (ii) non-proportional habitat-oriented sampling (from here on denoted as NPHS). Composite samples were collected following the WFD compliant INAG benthic macroinvertebrate sampling protocol, an adaptation of the AQEM methodology (refer to Hering *et al.*, 2004) where major multiple habitats are sampled in proportion to their presence within a sampling reach (INAG, 2008). A PHS sample consists of six 'sampling units' taken from all microhabitat types at the sampling site with a share of at least 5% coverage. The six 'sampling units' must be distributed according to the 'share' of microhabitats. This procedure results in sampling approximately 1.25 m² of the streambed. 'Sampling units' (1 m long and 0.25 m wide each) were collected by kick sampling in proportion to the number and extent of the following inorganic and organic substrate (microhabitat) categories: boulders (from here

on referred to as blocks; >256 mm), cobbles (64–256 mm), gravel and pebbles (2–64 mm), sand, silt and clay (52 mm) and both particulate organic matter (POM) and macrophytes. The composite sample (i.e. the six kick subsamples) was then placed in a plastic bag and fixed *in situ* using 4% formaldehyde.

The NPHS-oriented sampling procedure, also based on six sampling units, was distributed first by the number of microhabitat types present at the sampling site (i.e. not in proportion to percentage area covered), thereby ensuring 0.25 m minimum area per microhabitat type. Following this approach, the remaining habitat sampling effort area/units were distributed according to percentage cover area of each microhabitat type. For example, if three microhabitats were present at a given sampling site, the first three 0.25-m subsamples would be made for each microhabitat, regardless of percentage representivity. Remaining sampling effort (the remaining 3 × 0.25 m subsamples) would then be distributed according to the proportion of each microhabitat type present.

The NPHS-oriented approach is based on the assumption that if all microhabitats were equally represented at a given sampling site, then a single 1 m long by 0.25 m wide (0.25 m²) sampling unit per microhabitat would be sufficient for representative classification of ecological status. Thus, under the NPHS-oriented sampling protocol, the minimum sampling area (1 m long by 0.25 m² wide sampling unit) was considered representative of actual habitat community composition.

In the laboratory, the benthic macroinvertebrate samples were washed, sieved, sorted and identified to family level using a stereo microscope following the INAG protocol (2008). All individuals were picked out of the sample; subsampling was not used. Some taxa ($n = 2$) were not identified beyond order from macrophyte and cobble habitats. A single sample with the low abundance (five individuals— $n = 4$ Ephemeroptera and $n = 1$ Diptera) was also excluded from further analyses,

Data preparation and analysis

Calculation of metrics. Ecological quality was assessed by calculating the Iberian Biological Monitoring Working Party (IBMWP; Alba-Tercedor and Sanchez-Ortega, 1988) and the 'Índice Portugues de macroinvertebrados do Sul' (IPTIs) or south Portugal macroinvertebrate biotic index (INAG, 2008). The IBMWP is derived from the presence/absence of macroinvertebrate families, based on their sensitivity to organic pollution. Developed by INAG as part of the European inter-calibration freshwater group exercise (INAG, 2008), the IPTIs is a multimetric index that comprises the following metrics:

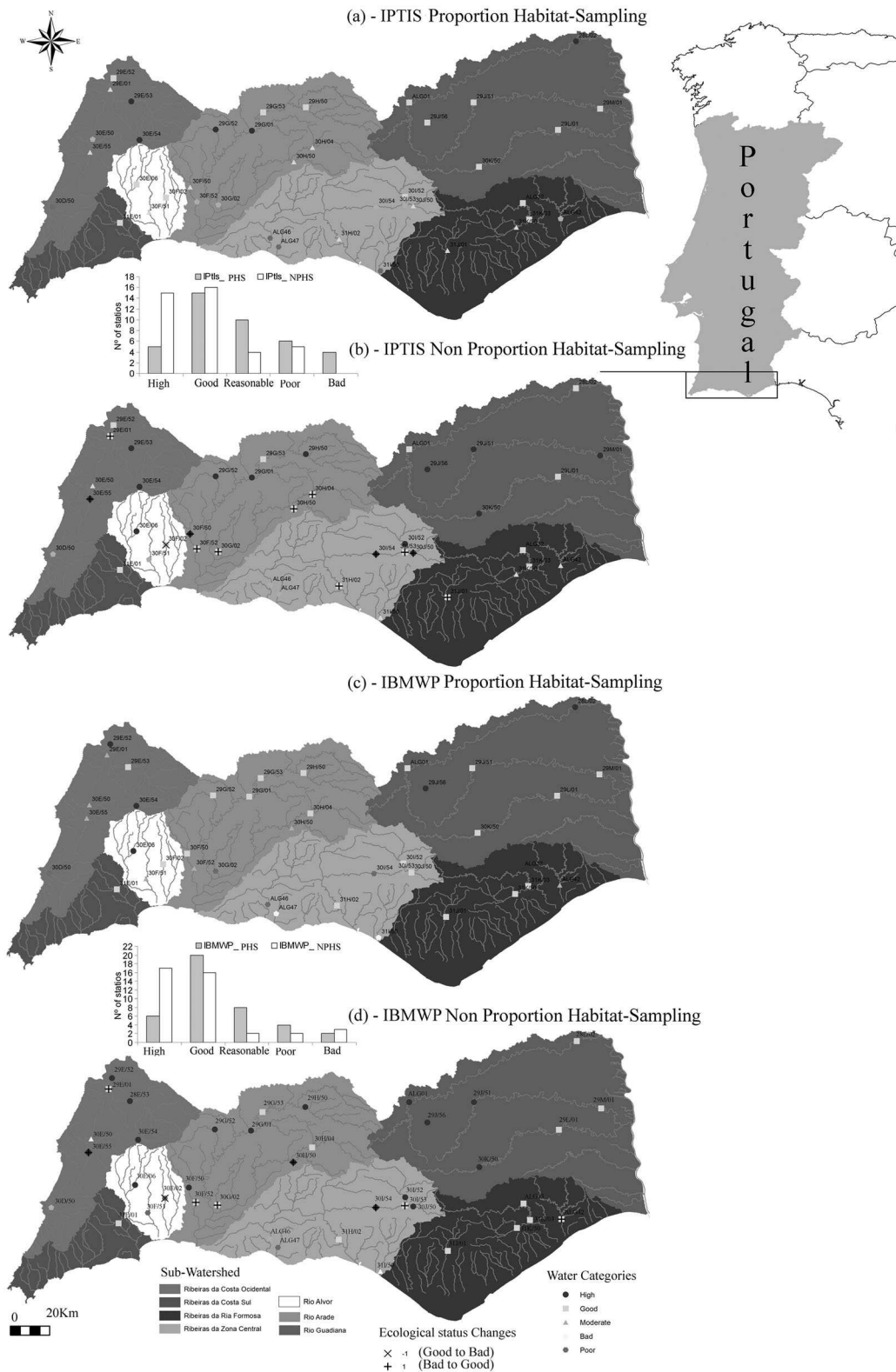


Figure 1. Map indicating the Algarve region of southern Portugal and the sampling sites and with IBMWP and IPTIS indices estimated for proportional habitat sampling (PHS) and non-proportional habitat sampling (NPHS) experiments. X – changes in ecological status from good to bad; and + changes in ecological status from bad to good.

$$\begin{aligned} \text{IPtIs} = & \text{No. of families} \times 0.4 + \text{EPT} \times 0.2 \\ & + (\text{IASPT} - 2) \times 0.2 + \text{Log} (\text{Sel. EPTCD} + 1) \\ & \times 0.2 \end{aligned}$$

where:

No. of families = the sum of the total number of macroinvertebrate families recorded; EPT = sum of the number of families belonging to the orders Ephemeroptera, Plecoptera and Trichoptera; IASPT = average score per taxon of the IBMWP, obtained by dividing the IBMWP sample score by the number of families in each sample that contribute to the score; Sel. EPTCD = sum of the number of individuals belonging to the Chloroperlidae, Nemouridae, Leptophlebiidae, Ephemerellidae, Philipotamidae, Elmidae, Leuctridae, Limnephilidae, Sericostomatidae, Dryopidae and Athericidae families.

The IPtIs ecological quality ratios (EQRs) were calculated based on observed and reference condition values for WFD-defined typologies based on the WFD reference condition approach for each typology in the Algarve region, (refer to Table I, EU 2008). Estimated values were used to classify sampling sites into one of four possible categories based on class boundary values (high/good, good/moderate, moderate/poor; poor/bad). We grouped sites assigned with high or good ecological status as good ecological status while sites with moderate, poor and bad ecological status were grouped together as having bad ecological status.

The percentage area sampled for each microhabitat type, the number of families and Shannon Weiner diversity (H') were determined for each sample collected following the different sampling protocols (sampled microhabitat types were grouped for the calculation of metrics). Statistical differences between composite metric values (IBMWP, EPT, IASPT, EPTCD, number of families, Shannon Weiner diversity), the IPtIs and the proportion of each microhabitat

covered for each sampling methodology were compared using the Wilcoxon sign sum rank matched pairwise test (Zar, 1996). The Wilcoxon test was used to compare differences in ecological classification obtained using the different sampling protocols (categorical data analysis). Pairwise comparison tests on the IPtIs component metrics were made on raw data (data was not normalized, and observed values were compared). Differences in community structure among pairwise samples of composite and habitat samples were examined by hierarchical clustering analysis of sample site presence/absence data (unweighted pair group average, Bray–Curtis similarity coefficient). Community structure overlap was compared between sampling methods using pairwise site similarity values.

Decision trees. Decision tree analyses were undertaken to compare and evaluate the effect of the different sampling protocols on IBMWP and IPtIs macroinvertebrate metric values and changes to classification of ecological status. Decision tree output identifies the conditions that lead to the tested sampling methods giving either the same or different index values, which may result in different classifications of ecological status. Decision tree analyses belong to a group of machine learning algorithms used for solving classification problems. Given a set of examples (data), where each example is comprised of attributes (descriptors or independent variables) and a class (outcome or target variable), the decision tree algorithm derives generic rules (or a model) according to which the class is expressed as a function of the attribute values.

Decision tree models have a tree-like structure, composed of nodes and branches that are interpreted as 'IF THEN' rules. Each tree comprises three types of nodes: the root (the starting node) at the top of the tree, internal nodes, and the terminal nodes or leaves containing the predictions of the class (target) values. Figure 6(a) gives an example of a decision tree, which is interpreted as follows:

Table I. Medians of reference condition values (RVs) for component metrics of the IPtIs and class boundary values for ecological status (high, good, moderate, poor and bad) for Algarve river typologies (INAG, 2009; EU 2008)

	Reference values					High/good boundary	Good/moderate boundary	Moderate/poor boundary	Poor/bad boundary
	EPT	Number of families	IASPT-2	Log (Sel. EPTCD + 1)	IPtIs	IPtIs	IPtIs	IPtIs	IPtIs
Small southern rivers	10	27	3.29	2.48	0.99	0.95	0.7	0.47	0.23
Medium-large southern rivers	9	21	3.37	2.57	0.98	0.97	0.72	0.48	0.24
Mountainous southern rivers	10.5	26	3.73	—	0.99	0.82	0.56	0.38	0.19
Calcareous rivers (Algarve)	10	27	3.29	2.48	0.99	0.95	0.7	0.47	0.23

IF (*attribute1_value* > *value1* and *attribute2_value* > *value2*) THEN (*target_value* = *value5*). IF (*attribute1_value* > *value1* and *attribute2_value* < *value2*) THEN (*target_value* = *value4*). IF (*attribute1_value* < *value1*) THEN (*target_value* = *value3*)

where the values of *attribute1* (root node), *attribute2* (internal node), and class (leaves) are measured values in the data set, *value 1* and *value 2* are the splitting values of attributes 1 and 2 respectively, and *value 3*, *value 4* and *value 5* are predicted values of the class variable.

In this study, the target variable was the difference in ecological status according to the type of sampling protocol applied. A target variable of zero meant that both protocols resulted in the same ecological status for the index under analysis. We employed the J48 algorithm, which is based on the top-down induction (TDI) decision tree principle embedded in the software package WEKA. After the tree is constructed from the training (learning) data set, the quality of the model (i.e. the accuracy of prediction) is assessed either by (i) simulating the model on a training data set or (ii) a test data set. The predicted values of the target are then compared with the actual values, and model accuracy is expressed as the percentage of correctly classified examples.

The data set comprised the microhabitat sampled (BLOCKS, COBBLE, GRAVEL, SAND, MACROPHYTES and POM) as attributes (explanatory variables) and ecological status (IBMWP and IPTIS) represented the class (target) variables or outcomes of each sample. Explanatory variables were pre-processed as follows:

- (1) Differences (DIF) among the area of each microhabitat type sampled were estimated by subtracting NPHS percentage cover from PHS percentage cover area ($\%DIF_i = NPHS\% - PHS\%$ for a given habitat type).
- (2) Data were min–max normalized (0 and 100) to standardize the scale within habitat station and among sites for each microhabitat.
- (3) Normalized percentage data were assigned into the following three classes: class 1 0–35% (susceptible to low levels of change), class 2 36–65% (susceptible to medium levels of change) and class 3 66–100% (susceptible to high levels of change). Minus and plus signs attributed to the derived values indicate either an increase or decrease in the sampled habitat area between the different sampling methodologies (NPHS vs PHS).

Class or response variables were also pre-processed. Our decision tree analysis response variables correspond to the difference in ecological status categories based on the metrics calculated (DIBMWP and DIPTIS) for each sampling method. Each sampling site was scored from 1 to 5 (bad

status = 1, poor = 2, moderate = 3, good = 4 and high = 5). For example, if the ecological status for a given site was given as high based on the NPHS-based sampling methodology (5) but good (4) for the PHS sampling methodology, quality status increased +1 in one ecological category (NPHS PHS: 5 – 4 = +1) in favour to NPHS field sampling.

RESULTS

Ecological and biotic indices

The number of families collected (Wilcoxon: $n = 40$; $Z = 3.36$; $p < 0.01$) and Shannon Wiener diversity (Wilcoxon: $n = 40$; $Z = 3.36$; $p < 0.01$) differed significantly between sampling protocols at each sampling site. The number of families and most diversity index values were higher following the NPHS-based methodology (Figure 2). With the exception of EPTCD, the values of the metrics used to derive the IPTIs (EPT and IASPT, refer to Figure 3) were also significantly higher based on the NPHS-oriented sampling method (Wilcoxon_{EPT}: $n = 40$; $Z = 3.36$; $p < 0.01$; Wilcoxon_{IASPT}: $n = 40$; $Z = 3.36$; $p < 0.01$).

The biological indices (IBMWP and IPTIs) and final ecological status (Figures 1 and 4) were typically higher for the NPHS-oriented sampling protocol. Wilcoxon test results indicated significant differences between the two sampling methods for both IPTIs (Wilcoxon: $n = 40$; $Z = 4.82$; $p < 0.01$) and IBMWP values (Wilcoxon: $n = 40$; $Z = 3.49$; $p < 0.01$). Categorical data analysis (based on classes) also indicated significant differences for both IPTIs (Wilcoxon: $n = 40$; $Z = 3.75$; $p < 0.01$) and IBMWP (Wilcoxon: $n = 40$; $Z = 2.87$; $p < 0.01$).

Changes in IPTIs values between PHS and NPHS sampling protocols resulted in different classifications of ecological status (Table II; Figure 1). A total of 50 and 27.5% of sites were classified as having bad ecological status based on the PHS and NPHS sampling protocols respectively. For IPTIs, a total of 29 sampling sites (72.5%) had different classifications of ecological status depending on the sampling protocol that was applied (Table II; Figure 1). A total of 26 sampling sites increased IPTIs-based ecological status applying the NPHS sampling protocol. A total of 12 of the 26 sites (46.2%) changed from bad to good ecological status. More precisely, eight sites classified as moderate ecological status ($n = 3$) changed to high (30E/55, 30F/50, 30J/50) or good ($n = 5$) (29E/01, 30H/04, 30H/50, 31H/02, 31J/01), and four sites classified as bad ecological status changed to good (30I/53, 30F/52, 30G/02) or high (30I/54). A single sampling site (30F/02) obtained lower ecological classification from good to poor based on the NPHS sampling protocol (Figure 1; Table II). The percentage of sites classified as having bad ecological status based on IBMWP scores for PHS and NPHS sampling methods were

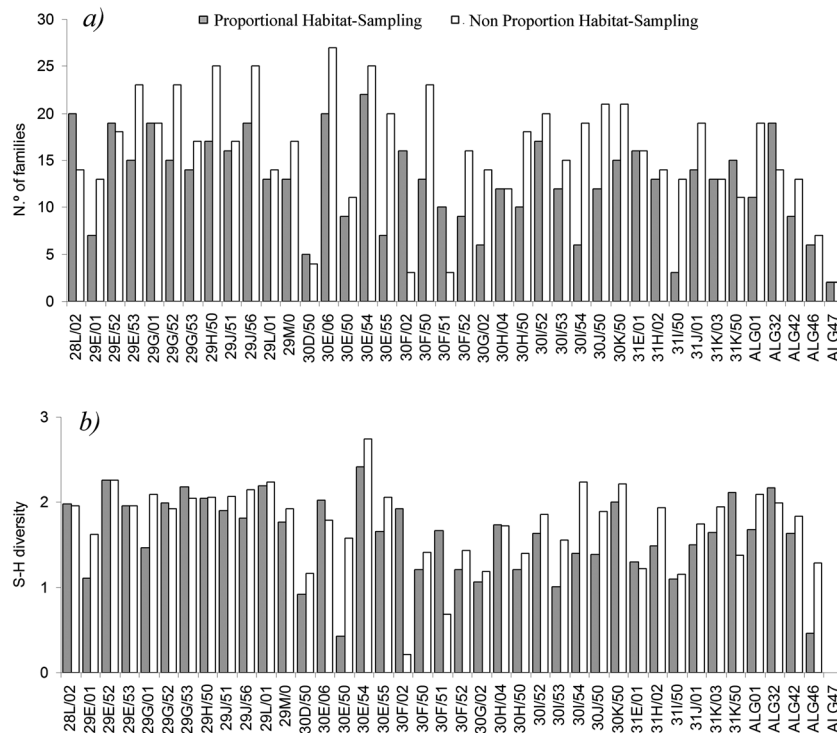


Figure 2. The number of families (a) and Shannon Weiner diversity (b) from the sites sampled ($n=40$) using proportional habitat sampling (PHS) and non-proportional habitat sampling (NPHS) sampling protocols.

35 and 17.5% respectively (Table II; Figure 1). A total of 24 sites (60%) obtained different ecological status classifications as a result of the different sampling protocols. Half of the sites (20 out of 40) had better ecological status following the NPHS sampling protocol. Eight sites improved from bad to good ecological status; six sites improved from moderate to good ecological status ($n=4$, 29E/01, 30F/52, 30I/53, ALG42) or high ($n=2$, 30E/55, 30H/50) and two sites from poor to high ($n=1$, 30I/54) or good ($n=1$, 30G/02). A single site dropped from good to poor ecological status based on the NPHS sampling protocol.

Cluster analysis revealed that many of the samples collected from the same site using the different protocols belonged to different end groups reflecting high levels of variability in benthic macroinvertebrate community composition using the different sampling protocols (Figure 5(a)). Only nine of the pairwise compared sites ($n=9$) shared similarity values greater than 63% (Figure 5(b)). The most commonly sampled microhabitats were cobble (50%), gravel (33%) and macrophytes (21%) respectively (Figure 6). Overall frequency occurrence of microhabitats in descending order was cobble (90%), macrophytes (75%), gravel (53%), sand (43%), blocks (30%) and POM (30%) (Figure 6(b)).

Differences in the percentage of the area covered of microhabitats employing the two sampling protocols (NPHS PHS) are given in Figure 6(a). Positive y-axis values

indicate an increase in the percentage of the area sampled for a specific microhabitat under the NPHS-based protocol. The most notable difference between the microhabitat types sampled based on the different sampling protocols is the significant increase in the macrophyte microhabitat type ($W=325$; $Z=3.34$; $p<0.01$) and POM ($W=58$; $Z=2.29$; $p<0.03$) following the NPHS-based protocol (Figure 6). Samples taken using the PHS sampling protocol comprised predominately cobble microhabitats, the most representative microhabitat type sampled based on this approach. The increase in macrophytes and POM microhabitat area using the NPHS-based protocol led to a significant reduction in the amount of cobble microhabitats surveyed ($W=-44$; $Z=-3.99$; $p<0.01$) based on the two protocols. These changes in the area of microhabitats sampled appear to be the primary driver behind changes in ecological status derived from the tested metrics.

Decision tree models

The decision tree model for the IBMWP results (model accuracy 55%; Figure 7(b)) can be interpreted as follows: If macrophyte cover decreases ($\text{MACROPH} \leq -1$), then, no change is expected in the derived IBMWP score, that is, both sampling methods will result in the same ecological classification of status. If macrophyte cover increases

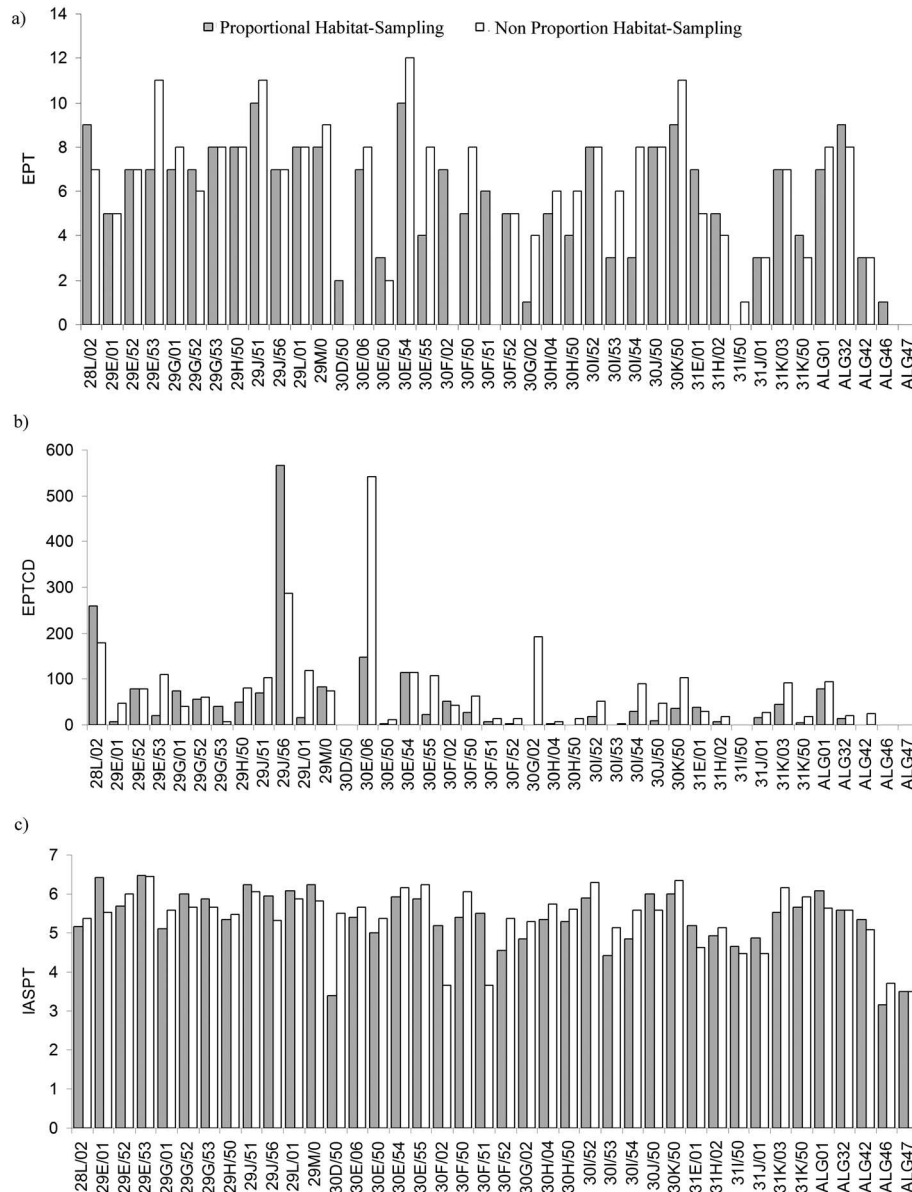


Figure 3. The number of (a) EPT families, (b) EPTCD family abundances and (c) IASPT metric scores collected using the proportional habitat sampling (grey bars) and non-proportional habitatsampling (white bars) oriented sampling protocols at sampling sites ($n = 40$).

($\text{MACROPH} > -1$), then, the presence of sand microhabitat (SAND) becomes important. An increase in sand microhabitat will result in higher ecological status if NPHS-based sampling is applied. However, if both sand and block microhabitats decrease, then, the difference in the sampling protocols is 1 (left-hand side of the model). If the block microhabitat slightly increases, that is, if ($\text{MACROPH} > -1$) and ($\text{SAND} < 1$) and ($-1 < \text{BLOCKS} \leq 1$), then, the presence of the POM microhabitat becomes the principal driver in the difference in values derived from the two sampling protocols. If POM increases significantly ($\text{POM} > 1$), then, ecological status will increase; otherwise, no change in

IBMWP values derived from the two sampling methodologies will be expected. Decision tree model accuracy for IPTIs (57.5%, Figure 7(c)) was similar to the IBMWP decision tree model. The right-hand side of the decision trees ($\text{COBBLE} > -2$) shows that based on the NPHS-based protocol, a slight decrease (-1) or increase ($1, 2, 3$) in COBBLE microhabitat results in an increase in ecological status if there is an increase in COBBLE and GRAVEL microhabitats, regardless of the presence of macrophyte microhabitats. The magnitude of the increase in the index value will then depend on the proportion of cobble microhabitat present.

HABITAT-ORIENTED SAMPLING

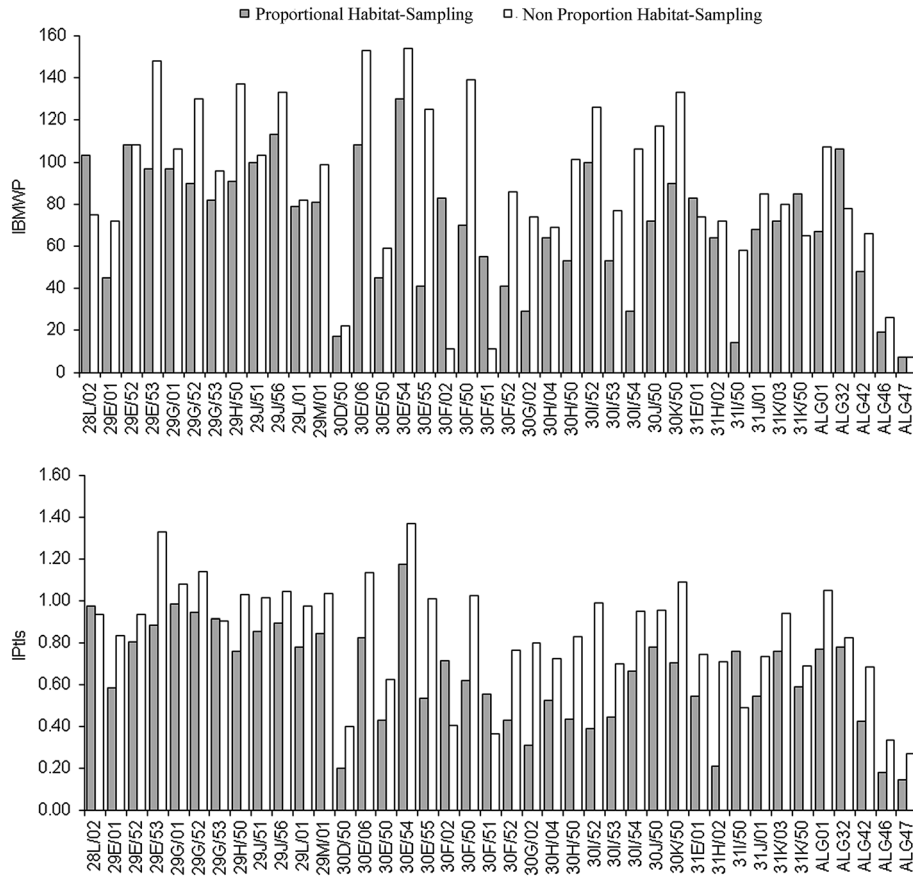


Figure 4. IPTiS and IBMWP metric values collected using the proportional habitat sampling (grey bars) and non-proportional habitat sampling (white bars) oriented sampling protocols at sampling sites ($n = 40$).

Results clearly show that the greatest disparity between derived values using the two different sampling protocols occurred when the proportion of macrophyte microhabitat

increased and cobble microhabitat decreased. These findings emphasize the importance of the presence of macrophyte microhabitats for increasing ecological status, although the

Table II. A summary of the number of sites in each ecological classification (EQS) for the multimetric IPTIs index and the IBMWP score

IPTIs	Number of stations (%)				WFD classification			Sampling differences (PHS – NPHS)	
	PHS	NPHS	PHS (%)	NPHS (%)	EQSt	PHS (%)	NPHS (%)	Increase EQSt (good)	Decrease EQSt (bad)
High	5	15	12.5	37.5	Good	50	72.5	26 (12)	3 (1)
Good	15	14	37.5	35					
Moderate	10	5	25	12.5	Bad	50	27.5		
Poor	6	4	15	10					
Bad	4	0	10	0					
IBMWP									
High	6	17	15	42.5	Good	65	82.5	19 (8)	4 (1)
Good	20	16	50	40					
Moderate	8	2	20	5	Bad	35	17.5		
Poor	4	3							
Bad	2	3	15	12.5					

The column ‘Sampling differences’ lists the number of sites that change in EQS as a result of the habitat-oriented sampling protocol and those that have changed EQS (in parentheses) based on Water Frame Directive classification of ecological status (good or bad). PHS, proportional habitat sampling; NPHS, non-proportional habitat sampling

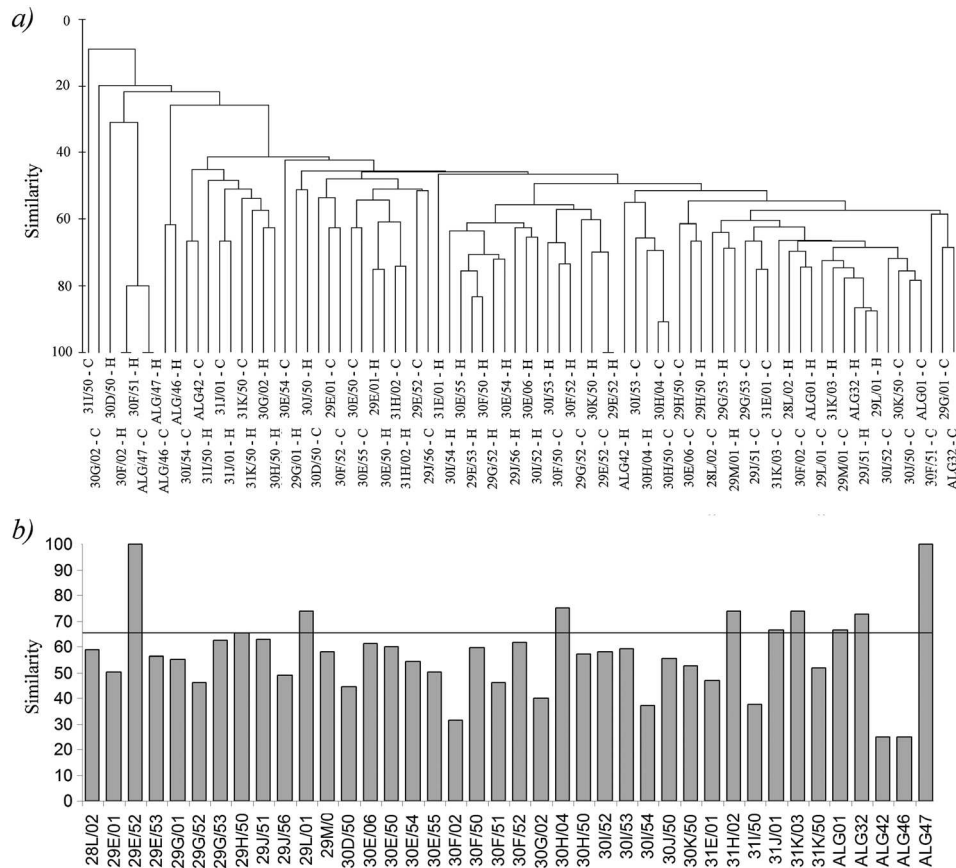


Figure 5. Bray Curtis distance based cluster analysis (a) comparing community structure among sampling sites for the proportional habitat sampling and non-proportional habitat sampling-oriented field sampling procedures and (b) corresponding percent similarity within sampling sites based on the two sampling methods.

degree of increase is reduced when both macrophyte and cobble microhabitats are present (refer to the bottom of the right-hand side of the IPTIs decision tree model).

DISCUSSION

River systems characterized by temporary flow regimes are relatively common, highly dynamic and will probably become more widespread as a result of climate change (Larned *et al.*, 2010). Aquatic communities inhabiting temporary river systems are subject to seasonal and inter-annual cycles of flow, drought and drying. These factors strongly influence the availability of habitats that can be potentially occupied as well as community structure and function (Hughes *et al.*, 2009; Hughes *et al.*, 2010; Gallart *et al.*, 2012). Reliable, standardized field methodologies and in-depth knowledge of how hydrology and ecological processes are coupled in these systems are vital for the estimation of WFD compliant ecological status (Bonada *et al.*, 2006; Gallart *et al.*, 2012; Tagliaferro *et al.*, 2013). The two sampling protocols

tested in this study produced differing results, which should be considered for WFD planning and management of Mediterranean temporary stream systems because ecological status is a key to the implementation of the WFD (Gallart *et al.*, 2012).

Although the sampling effort is identical for both field protocols, they differed essentially in the area per microhabitat covered. Both metric values and derived ecological status tended to increase when the NPHS was applied. Wilcoxon statistical analysis showed that significant changes occurred when macrophytes and POM microhabitat types were included in kick sampling areas using the NPHS-based sampling protocol. Results from the decision tree models clearly indicated that changes in the presence and extent of specifically organic (macrophytes and POM) but also inorganic microhabitats sampled (cobble) using either the NPHS based or PHS sampling methods significantly influence metric values and the resulting ecological status classification. Moreover, results showed that one microhabitat could be important depending on the availability of other microhabitats and that slight increases (or decreases) in the area of some

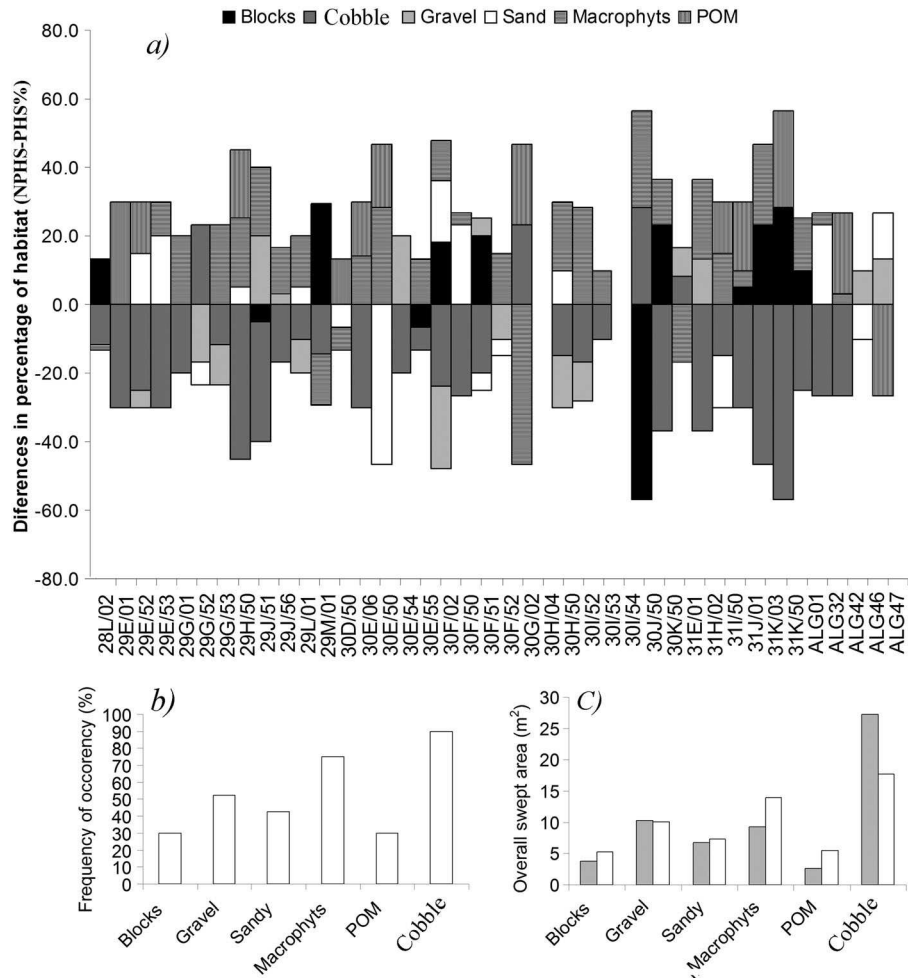


Figure 6. (a) Percentage differences between the area sampled using proportional habitat sampling (PHS) and non-proportional habitat sampling (NPHS) invertebrate sampling protocols. (b) Frequency occurrence of each microhabitat and (c) overall sampled area for proportional habitat sampling (grey) and non-proportional habitat sampling (white) sampling methods.

microhabitats could significantly affect the resulting scores and ecological status for both IBMWP and IPTIS.

A fundamental tenet of ecology is how space and habitat selectivity requirements influence community structure. Particular macroinvertebrate traits associated with a given habitat (morphological adaptations, feeding behaviour and available food resources, reproductive strategies) are related to evolutionary adaptations resulting in feeding, reproduction and growth strategies that best exploit available resources within given habitats (Southwood, 1977, 1988). Mediterranean intermittent rivers, such as those in the Algarve region, undergo strong annual and inter-annual variation in flow, a phenomenon directly related to precipitation patterns influenced by different phases of the North Atlantic Oscillation (Hughes *et al.*, 2012; Trigo *et al.*, 2004). Such variation strongly influences the spatial and temporal hydrological regime of flooding and subsequent drought/drying in Mediterranean fluvial systems, biotic and abiotic processes

such as sediment transport and patterns of erosion and deposition (Gasith and Resh, 1999; Roy *et al.* 2003; Hooke, 2006; Reid *et al.*, 2010). These environmental factors affect local habitat availability, that is, mesohabitats such as riffles and pools and associated microhabitats, determining the available resources and the structure and function of biological communities that occupy them (Magalhães *et al.*, 2002; Bernardo *et al.*, 2003; Bêche and Resh 2007a, 2007b; Bonada *et al.* 2007a, 2007b; Díaz *et al.*, 2008; Hughes *et al.*, 2009; Hughes *et al.*, 2010). It is the dynamic regime of intermittent rivers that lead to adaptive shifts in communities (Munné and Prat, 2011).

Although many benthic freshwater species seem to be generalist in terms of the type of habitat they occupy, some species display preferences for specific types of habitat, often associated with food availability, species feeding behaviour and morphology (Giller and Malmqvist, 1998). Specific substrate properties influence the distribution of

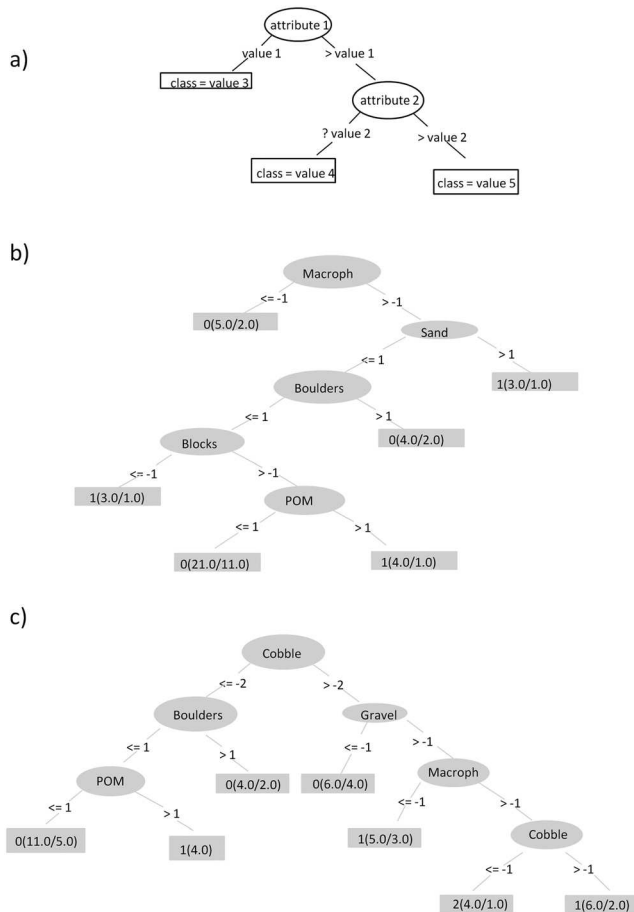


Figure 7. Diagrams of decision tree models developed to test how proportional habitat sampling and non-proportional habitat sampling methods affect derivation of ecological status classification in relation to microhabitats sampled. Decision tree (a) illustrates how decision model trees are developed; (b) a decision tree for the IBMWP score and (c) a decision tree for the multimetric IPTIs index.

macroinvertebrates (Thomaz and Cunha, 2010). For example, inorganic sandy substrates are generally considered as unstable habitats with little organic matter and therefore tend to support low biotic diversity. Decision tree model results in this study showed that an increase in sand microhabitat type led to an increase in IBMWP-based ecological status (Figure 6) when associated with the presence of macrophyte microhabitats. Macrophytes are an important organic microhabitat, providing shelter, food and habitat complexity as well as influencing biotic interactions (Thomaz and Cunha, 2010). Emergent macrophytes and surrounding riparian vegetation are also important for the airborne adult phase of many aquatic insects (Abelho, 1996; Wallace and Anderson, 1996). Vegetation in pool areas in particular also plays an important role by retaining valuable resources such as POM during periods of flood (Armitage *et al.*, 2001). Samples collected

during this study support these observations because POM habitat types primarily occurred close to areas of macrophytes. The decision tree models provide important information regarding how microhabitat type and presence can affect the values of metrics. The results of decision tree models showed that POM availability drove changes in ecological status and indicated that distribution patterns of key macroinvertebrate indicator species were associated with the distribution of this organic microhabitat. Similarly, the presence of POM was found to significantly influence the elevated abundance of some species of Ephemeroptera nymphs using *in situ* experiments of implanted substrata of different inorganic microhabitats similar to those examined in this study (sand of 1.5 mm mean diameter; pebbles of 39.6 mm mean diameter; cobbles of 67.5 mm mean and small small rocks; Hearnden and Pearson, 1991). These results clearly indicate partitioning of the available habitat and associated resources.

Results of the cluster analyses (based on presence/absence data) showed that most community similarity levels were below 63%, indicating marked differences in the area of microhabitat sampled using the different sampling protocols and potential niche partitioning of indicator organisms. This would affect presence/absence of species (~37% dissimilarity) in the samples collected using the different protocols (Figure 5 (a)) and would ultimately be reflected in the determination of ecological status. The results of this study support our hypothesis that when a given microhabitat is sampled according to the proportion of its presence at a sampling site (i.e. composite sampling), the area sampled may not capture the full potential diversity distributed across the multiple types of the patches. Results of pairwise tests showed that the number of families and diversity was significantly lower in PHS samples collected in this study. Although the WFD compliant NPHS composite sampling-based protocol developed for Portugal (INAG, 2008) states that less represented habitats situated close to dominant habitats should be also sampled, the minimum amount to sample is not specified. As a result, less frequently occurring habitats are sampled according to their proportion of their occurrence at a given site. There appear to be very few studies on the effect of under-represented lotic habitats and the influence of additional taxa in biological assessment. However, Haase *et al.* (2009) also found that multimetric assessment results were affected by sampling 'additional' microhabitats with low coverage across 48 sampling sites in Germany using the STAR/AQEM sampling protocols. Interestingly, Bonada *et al.* (2006) compared two different macroinvertebrate sampling methodologies in Mediterranean stream systems on two different continents and found that they produced very similar results, expressed via metrics. These results indicated that sampling a single habitat may be sufficient for assessing water quality, but a multihabitat sampling approach to sampling is recommended where intrinsic variability of macroinvertebrate assemblages

is high (e.g. in undisturbed sites in regions with Mediterranean climates). However, in the work of Bonada *et al.* (2006), the sampling methods varied in terms of equipment used, the process of sampling, sorting procedures and also segregation of habitats, while in the current study, all procedures were kept constant. In the present study, the similarity among sampling sites for the different sampling methods varied considerably. Therefore, based on the research of Bonada *et al.* (2006) and the present study, a multihabitat sampling approach should be used for RBPs in Mediterranean streams.

Based on the findings of this study, the WFD composite sampling method (PHS) probably omits taxa occupying microhabitats occurring in very low proportions, that is, areas that may not be sampled in some protocols as a result of the limited area. This is a very important consideration for the ecological assessment of Mediterranean intermittent systems. Current WFD assessment of ecological status in intermittent or temporary streams is only possible when flow conditions result in sufficient diversity and availability of habitats for biomonitoring to be undertaken (Munné and Prat, 2009, 2011). This is probably the best explanation for the changes in ecological status trend based on the NPHS sampling protocol. A single sampling site (Table II) showed a reduction in ecological status to bad when based on the NPHS-based sampling protocol compared with the WFD (NPHS) sampling protocol. POM microhabitats were absent from this sampling site, and the gravel and cobble microhabitat occupied 88% of the area sampled. We recommend that in-depth studies on the associations between macroinvertebrate fauna and habitat-type interactions be carried out to provide information on niche availability and change in the values of metrics or indices as a result of the different sampling methods. García-Roger *et al.* (2011) showed that seasonal changes in habitat features influence aquatic macroinvertebrate assemblages in Mediterranean streams and that they were related to mesohabitats and microhabitats, particularly mineral and organic microhabitat patches. Differences recorded were because of the loss of taxa from some habitats and individual taxa displaying strong habitat affinities. Despite the long history of ecological studies on temporary streams (Stehr and Branson, 1938; Hynes, 1958), there is still a paucity of knowledge regarding the factors influencing the structure and composition of aquatic macroinvertebrate communities in 'temporary seasonal' streams. However, our results show a clear increase environmental quality status and classification with NPHS. Exactly how changes in the sampling protocol leads to the changes in community structure recorded (e.g. families involved and distribution of families within each habitat) should be analysed in further detailed hydroecological studies in Portuguese intermittent systems.

A method that could be applied to Algarve streams is the Mediterranean Intermittent River Management (MIRAGE) project approach that has examined hydrological,

biogeochemical and sediment transport processes in European temporary Mediterranean river systems. These studies provide a detailed baseline characterization that will guide knowledge for improved assessment and management of these temporary lotic systems (Gallart *et al.*, 2012). Four temporary regimes (permanent, temporary pools, temporary dry and episodic) have already been identified from studies carried out in several temporary Mediterranean river systems as part of this project. The application of MIRAGE methodologies to the temporary streams and rivers of the Algarve region would help to address questions concerning the characterization and ecological assessment of temporary systems raised by the findings of this study. For example, discharge records from gauging stations or the development of rainfall-run-off models (Hughes *et al.*, 2012) is vital for simulating spatiotemporal patterns for the four defined aquatic states and developing studies to obtain the detailed knowledge of the biological conditions associated with each aquatic state. Once appropriate baseline data is obtained, multivariate statistical analyses can be used to assess predictive multivariate or multimetric ecological assessment techniques that attempt to explore the spatiotemporal changes in habitat availability and community structure in these systems. In the meantime, we advocate that the present sampling protocol should be modified to include the tested NPHS approach sampling method tested and that the findings of this study help to contribute to the development of more robust assessment methods for Portugal's intermittent river systems.

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Supporting information

Additional supporting information may be found in the online version of this article at the publisher's web site.

Supplement 1. List of taxa (families) with average abundances (m²) for proportional habitat sampling (P HS) and non-proportional habitat sampling (NP HS) sampling methods.